



SupplyGuard

— PREDICT. PROTECT. DELIVER. —

Delivery Delay Risk Prediction for E-Commerce Operations

An end-to-end analytics and machine learning project to monitor delivery performance and prioritize high-risk orders.

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The Problem:

Late deliveries create operational pressure and damage customer satisfaction

E-commerce companies need to know:

- Where late deliveries are concentrated
- Which order profiles are more exposed to delay
- Which orders need earlier operational attention



Can we identify orders with higher late-delivery risk before the customer is affected?

Project Approach:

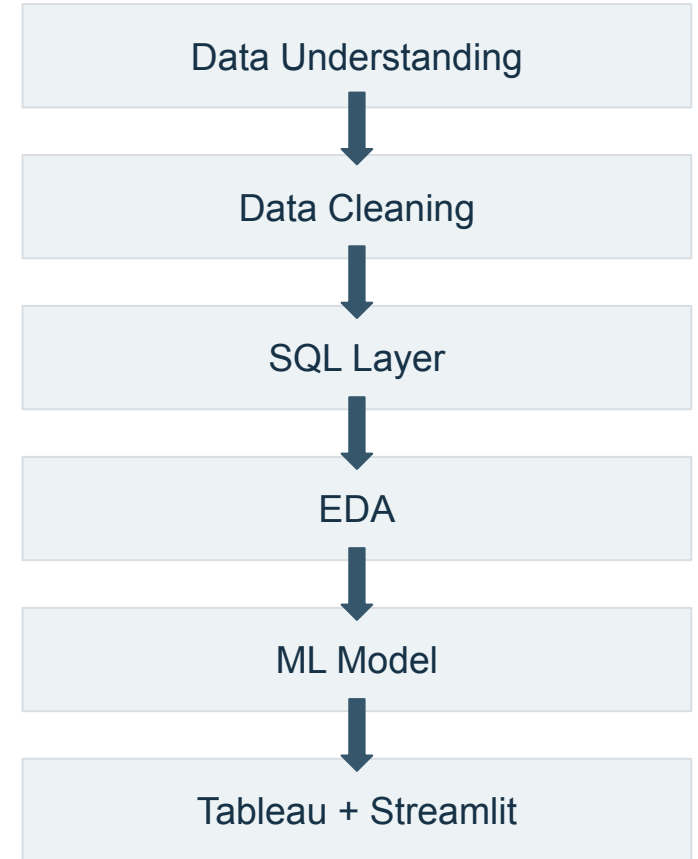
From raw data to operational decision support

The project follows an end-to-end analytics workflow:

1. Prepare and structure the [data](#)
2. Analyze delivery performance and risk patterns
3. Build a leakage-safe prediction model
4. Deliver business tools through Tableau and Streamlit

Main principle

Predict late-delivery risk using only pre-delivery information.



Target Definition: What counts as a late delivery?

Late delivery was defined at calendar-date level:

```
delivered_date > estimated_delivery_date
```

Orders delivered on the estimated delivery date are not considered late.

Why this matters:

The estimated delivery field represents a promised delivery date, not an exact timestamp deadline.

Timestamp-based logic would have incorrectly classified 1,292 same-day deliveries as late.

Delivered orders analyzed

96,470

Late orders

6,534

Late delivery rate

6.77%

Delivery Performance Insights:

Late delivery is not random

Late delivery concentrates by geography, route type and peak periods, and clearly impacts customer satisfaction.

Same-state vs cross-state late rate

4.52% vs 8.04%

SP → RJ

14.17% late rate

8,131 orders

Peak-month late rate

15.15%

Review score impact (On-time vs late)

4.29 vs 2.27

From prediction to prioritization

The model converts delivery patterns into an operational risk-prioritization tool.

Final model:

- Random Forest
- Threshold: 0.16
- PR-AUC: 23.84%
- Recall: 34.20%

Flagged order share

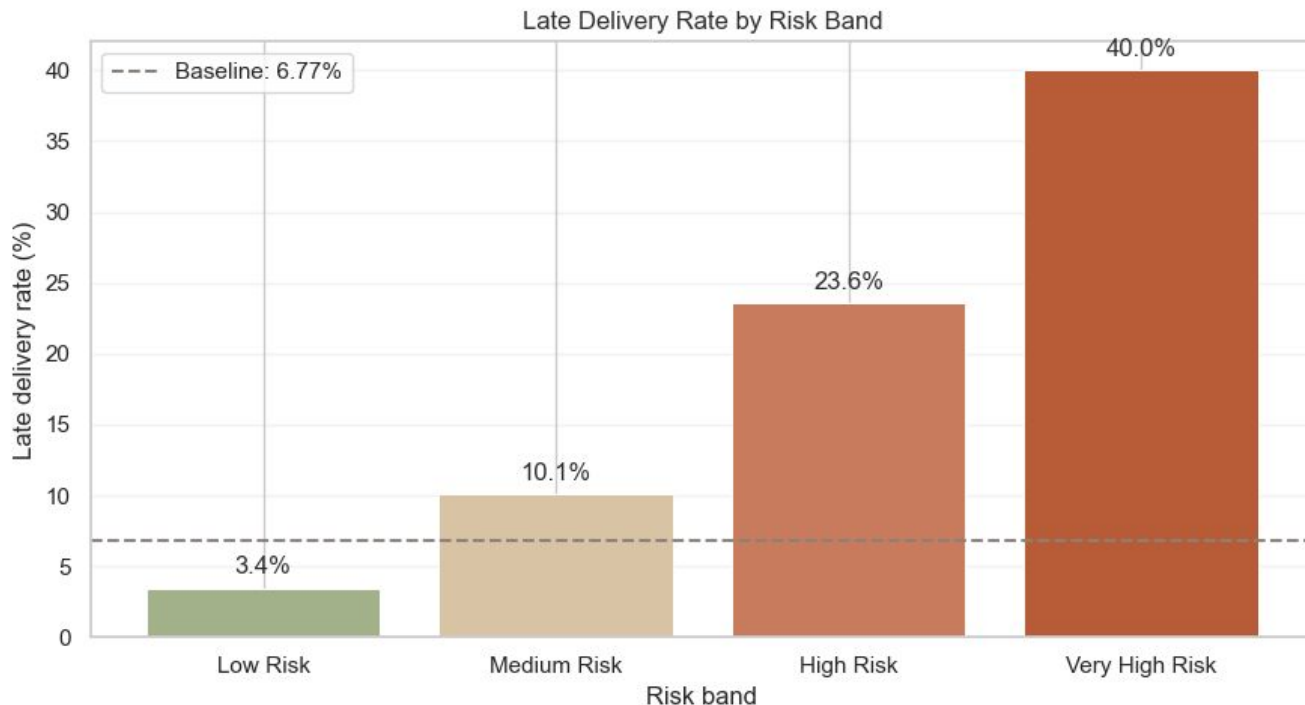
7.95%

Flagged late rate

29.14%

Lift vs baseline

4.3x



The model is not a perfect detector. It helps operations focus on orders with higher-than-average risk.

From Analysis to Business Tools

Tableau monitors performance. Streamlit supports order-level action.

Tableau Dashboard

Monitoring & reporting layer

- Historical delivery performance
- Geographic and route risk
- Risk-band monitoring
- Executive overview

[Open Tableau Dashboard](#)

Streamlit App

Operational scoring layer

- Score individual orders
- Upload batch orders
- Assign risk bands
- Flag high-risk deliveries
- Recommend action

[Open Streamlit App](#)



Thank You

Johannes Vidal

GitHub Repository:

<https://github.com/JohannesVidal/supplyguard-late-delivery-risk-prediction>

Questions?